Trigonometric Ratio: Questions and Full Solutions

Question 01

English

The value of $\displaystyle\sum_{k=1}^{45} \frac{1}{\cos 1^\circ + \cos(2k-1)^\circ}$ is equal to

(A) $\operatorname{cosec} 1^\circ$

(B) $\dfrac{1}{2}\cot 1^\circ$

(C) $\dfrac{1}{2}\operatorname{cosec} 1^\circ$

(D) $\dfrac{1}{2}(\operatorname{cosec} 1^\circ - \sec 1^\circ)$

Only one option is correct.

हिन्दी

$\displaystyle\sum_{k=1}^{45} \frac{1}{\cos 1^\circ + \cos(2k-1)^\circ}$ का मान बराबर है

(A) $\operatorname{cosec} 1^\circ$

(B) $\dfrac{1}{2}\cot 1^\circ$

(C) $\dfrac{1}{2}\operatorname{cosec} 1^\circ$

(D) $\dfrac{1}{2}(\operatorname{cosec} 1^\circ - \sec 1^\circ)$

केवल एक विकल्प सही है।

Solution

Use $\cos P + \cos Q = 2\cos\dfrac{P+Q}{2}\cos\dfrac{P-Q}{2}$ with $P = (2k-1)^\circ$, $Q = 1^\circ$: $\dfrac{P+Q}{2} = k^\circ$ and $\dfrac{P-Q}{2} = (k-1)^\circ$.

Hence $\cos 1^\circ + \cos(2k-1)^\circ = 2\cos k^\circ \cos(k-1)^\circ$. For $1 \le k \le 45$ both cosines are positive, so no term is undefined.

Since $\tan A - \tan B = \dfrac{\sin(A-B)}{\cos A \cos B}$, with $A = k^\circ$, $B = (k-1)^\circ$ we get $\dfrac{1}{\cos k^\circ \cos(k-1)^\circ} = \dfrac{\tan k^\circ - \tan(k-1)^\circ}{\sin 1^\circ}$.

Therefore the k-th term equals $\dfrac{\tan k^\circ - \tan(k-1)^\circ}{2\sin 1^\circ}$.

Adding for $k = 1, 2, \ldots, 45$, all intermediate tangents cancel: $\displaystyle\sum_{k=1}^{45}(\tan k^\circ - \tan(k-1)^\circ) = \tan 45^\circ - \tan 0^\circ = 1$.

So the sum is $\dfrac{1}{2\sin 1^\circ} = \dfrac{1}{2}\operatorname{cosec} 1^\circ$ (about 28.65).

Check of the distractors: forgetting the factor 2 gives (A); ending the telescoping at $\tan 1^\circ$ gives $\dfrac{1-\tan 1^\circ}{2\sin 1^\circ} = \dfrac{1}{2}(\operatorname{cosec} 1^\circ - \sec 1^\circ)$, option (D).

Correct Option: (C)

Question 02

English

For each positive integer n, let $S_n = \sin 6^\circ + \sin 12^\circ + \sin 18^\circ + \cdots + \sin (6n)^\circ$. Then the value of $\sum_{n=1}^{29} \frac{1}{S_n}$ is equal to

(A) $\cot 3^\circ$
(B) $\cot 3^\circ \operatorname{cosec} 3^\circ$
(C) $\cot 6^\circ$
(D) $\operatorname{cosec} 3^\circ$

Only one option is correct.

हिन्दी

प्रत्येक धनात्मक पूर्णांक n के लिए माना कि $S_n = \sin 6^\circ + \sin 12^\circ + \sin 18^\circ + \cdots + \sin (6n)^\circ$ है। तब $\sum_{n=1}^{29} \frac{1}{S_n}$ का मान बराबर है

(A) $\cot 3^\circ$
(B) $\cot 3^\circ \operatorname{cosec} 3^\circ$
(C) $\cot 6^\circ$
(D) $\operatorname{cosec} 3^\circ$

केवल एक विकल्प सही है।

Solution

Closed form of S_n (angles in AP, common difference 6°). By the product-to-sum formula $2\sin A \sin B = \cos(A-B) - \cos(A+B)$:

$2\sin 3^\circ \sin(6k)^\circ = \cos(6k-3)^\circ - \cos(6k+3)^\circ$.

Adding for $k = 1$ to n, consecutive terms cancel: $2\sin 3^\circ \, S_n = \cos 3^\circ - \cos(6n+3)^\circ$.

Sum to product on the right: $\cos 3^\circ - \cos(6n+3)^\circ = 2\sin(3n+3)^\circ \sin(3n)^\circ$. Hence $S_n = \dfrac{\sin(3n)^\circ \sin(3n+3)^\circ}{\sin 3^\circ}$.

For $1 \le n \le 29$ the angles $3n^\circ$ and $(3n+3)^\circ$ lie in $(0^\circ, 90^\circ]$, so $S_n > 0$ and every term $\dfrac{1}{S_n}$ is defined.

Now $\cot x - \cot y = \dfrac{\sin(y-x)}{\sin x \sin y}$. With $x = (3n)^\circ$, $y = (3n+3)^\circ$: $\cot(3n)^\circ - \cot(3n+3)^\circ = \dfrac{\sin 3^\circ}{\sin(3n)^\circ \sin(3n+3)^\circ} = \dfrac{1}{S_n}$.

So $\sum_{n=1}^{29} \dfrac{1}{S_n} = \sum_{n=1}^{29} (\cot(3n)^\circ - \cot(3n+3)^\circ) = \cot 3^\circ - \cot 90^\circ = \cot 3^\circ$.

Keeping an extra factor $\dfrac{1}{\sin 3^\circ}$ by mistake gives (B); (C) and (D) are close numbers but wrong.

Correct Option: (A)

Question 03

English

Let A, B and C be the angles of a triangle ABC such that $A - B = \frac{\pi}{4}$ and $\tan C = 3$. Then $\tan A + \tan B$ is equal to

(A) $\frac{3\left(\sqrt{5}-1\right)}{2}$

(B) $\frac{3\left(\sqrt{5}+1\right)}{2}$

(C) $\frac{3+\sqrt{5}}{2}$

(D) $\frac{3\sqrt{5}}{2}$

Only one option is correct.

हिन्दी

माना कि A, B तथा C त्रिभुज ABC के कोण हैं, जिनके लिए $A - B = \frac{\pi}{4}$ तथा $\tan C = 3$ है। तब $\tan A + \tan B$ बराबर है

(A) $\frac{3\left(\sqrt{5}-1\right)}{2}$

(B) $\frac{3\left(\sqrt{5}+1\right)}{2}$

(C) $\frac{3+\sqrt{5}}{2}$

(D) $\frac{3\sqrt{5}}{2}$

केवल एक विकल्प सही है।

Solution

Since $\tan C = 3 > 0$, C is acute: $\cos C = \frac{1}{\sqrt{10}}$, $\sin C = \frac{3}{\sqrt{10}}$.

$A + B = \pi - C$, so $\cos(A+B) = -\cos C = -\frac{1}{\sqrt{10}}$. Also $\cos(A-B) = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$.

Compound angles: $\cos(A-B) + \cos(A+B) = 2\cos A\cos B$ and $\cos(A-B) - \cos(A+B) = 2\sin A\sin B$.

Hence $\tan A\tan B = \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{10}}}{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{10}}}$. Multiplying numerator and denominator by $\sqrt{10}$: $\frac{\sqrt{5}+1}{\sqrt{5}-1} = \frac{(\sqrt{5}+1)^2}{4} = \frac{3+\sqrt{5}}{2}$.

(Both A and B are acute: $A + B = \pi - C \approx 108.4^\circ$ and $A - B = 45^\circ$ give $A \approx 76.7^\circ$, $B \approx 31.7^\circ$, so the product is positive, as found.)

For angles of a triangle, $\tan(A+B) = -\tan C$ gives the conditional identity $\tan A + \tan B + \tan C = \tan A\tan B\tan C$.

So $\tan A + \tan B = \tan C\left(\tan A\tan B - 1\right) = 3\left(\frac{3+\sqrt{5}}{2} - 1\right) = \frac{3(1+\sqrt{5})}{2}$.

Option (C) is the product $\tan A\tan B$, a likely slip; (A) comes from the rejected root $\tan A\tan B = \frac{3-\sqrt{5}}{2}$.

Correct Option: (B)

Question 04

English

A sector OAB of a circle with centre O has arc AB of length 6π cm and area 54π cm^2. A circle Γ lies inside the sector and touches the radii OA, OB and the arc AB. If the area of the part of the sector lying outside Γ is $k\pi$ cm^2, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

केन्द्र O वाले एक वृत्त के त्रिज्यखण्ड OAB के चाप AB की लम्बाई 6π cm तथा क्षेत्रफल 54π cm^2 है। एक वृत्त Γ त्रिज्यखण्ड के अन्दर स्थित है तथा त्रिज्याओं OA, OB और चाप AB को स्पर्श करता है। यदि त्रिज्यखण्ड के उस भाग का क्षेत्रफल, जो Γ के बाहर है, $k\pi$ cm^2 है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For a sector of radius r and arc length ℓ, the area is $\frac{1}{2}r\ell$ (since area $= \frac{1}{2}r^2\theta$ and $\ell = r\theta$, with θ in radians).

So $54\pi = \frac{1}{2} \cdot r \cdot 6\pi$, which gives $r = 18$ cm. The central angle is $\theta = \frac{\ell}{r} = \frac{6\pi}{18} = \frac{\pi}{3}$ radian, that is $60°$.

A circle touching both radii has its centre P on the bisector of $\angle AOB$, which makes $30°$ with each radius. If ρ is its radius, the distance from P to OA is $OP \sin 30° = \rho$, so $OP = 2\rho$.

Touching the arc internally means $OP + \rho = r$: $3\rho = 18$, so $\rho = 6$ cm. Area of $\Gamma = 36\pi$ cm^2.

Area of the sector outside $\Gamma = 54\pi - 36\pi = 18\pi$ cm^2, so $k = 18$.

Final Answer: 18

Question 05

English

A ceiling fan has three blades whose tips P, Q, R are at a distance of 0.9 m from the axis, the angle between the lines from the axis to any two of the tips being $120°$, and Q, R follow P in the anticlockwise sense. Take the axis as the origin, the x-axis pointing east and the y-axis pointing north, with distances in metres. Initially P is at $(0.9, 0)$. The fan turns anticlockwise until the tip P has travelled a distance of 19.05π m along its circular path. Then the sum of the y-coordinates of those tips that lie below the x-axis is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक छत के पंखे की तीन पंखुड़ियों के सिरे P, Q, R अक्ष से 0.9 m की दूरी पर हैं, अक्ष से किन्हीं दो सिरों तक की रेखाओं के बीच का कोण $120°$ है, तथा वामावर्त दिशा में P के बाद Q, R आते हैं। अक्ष को मूलबिन्दु, x-अक्ष को पूर्व की ओर तथा y-अक्ष को उत्तर की ओर लीजिए, दूरियाँ मीटर में हैं। प्रारम्भ में P बिन्दु $(0.9, 0)$ पर है। पंखा वामावर्त घूमता है जब तक सिरा P अपने वृत्तीय पथ पर 19.05π m की दूरी तय नहीं कर लेता। तब x-अक्ष के नीचे स्थित सिरों के y-निर्देशांकों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Arc length $\ell = r\theta$ with θ in radians: $\theta = \dfrac{19.05\pi}{0.9} = \dfrac{127\pi}{6}$.

Since $\dfrac{127\pi}{6} = 10(2\pi) + \dfrac{7\pi}{6}$, after ten full turns P is at the angle $\dfrac{7\pi}{6}$ (that is $210°$) from the positive x-axis.

Q and R are $120°$ and $240°$ ahead of P anticlockwise: at $210° + 120° = 330°$ and $210° + 240° = 450°$, that is $90°$.

Allied angles: $\sin 210° = \sin(180° + 30°) = -\sin 30° = -\dfrac{1}{2}$ (third quadrant, sine negative); $\sin 330° = \sin(360° - 30°) = -\dfrac{1}{2}$ (fourth quadrant); $\sin 90° = 1$.

So $y_P = 0.9\left(-\dfrac{1}{2}\right) = -0.45$, $y_Q = -0.45$ and $y_R = 0.9$. Only P and Q are below the x-axis.

Required sum $= -0.45 + (-0.45) = -0.9$.

Final Answer: -0.9

Question 06

English

A particle moves anticlockwise on the circle $x^2+y^2 = 125^2$, starting from $A(125, 0)$. It travels an arc of length ℓ to reach a point B, and then a further arc of length 2ℓ to reach a point C, where ℓ is less than half the circumference of the circle. If the chord AB has length $50\sqrt{5}$, then the x-coordinate of C is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक कण वृत्त $x^2 + y^2 = 125^2$ पर बिन्दु $A(125, 0)$ से प्रारम्भ होकर वामावर्त गति करता है। यह ℓ लम्बाई का चाप तय करके बिन्दु B पर पहुँचता है, तथा फिर 2ℓ लम्बाई का और चाप तय करके बिन्दु C पर पहुँचता है, जहाँ ℓ वृत्त की परिधि के आधे से कम है। यदि जीवा AB की लम्बाई $50\sqrt{5}$ है, तो C का x-निर्देशांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Arc length = radius × angle (radian). With $\theta = \dfrac{\ell}{125}$, the arc AB subtends θ at O, and the arc BC of length 2ℓ subtends 2θ. Since $\ell < 125\pi$, $0 < \theta < \pi$.

So $B = (125\cos\theta, 125\sin\theta)$ and $C = (125\cos 3\theta, 125\sin 3\theta)$ (angles measured from OA).

Chord length: $AB = 2r\sin\dfrac{\theta}{2}$. Thus $250\sin\dfrac{\theta}{2} = 50\sqrt{5}$, so $\sin\dfrac{\theta}{2} = \dfrac{1}{\sqrt{5}}$; this fixes θ uniquely because $\dfrac{\theta}{2} \in \left(0, \dfrac{\pi}{2}\right)$.

Then $\cos\theta = 1 - 2\sin^2\dfrac{\theta}{2} = \dfrac{3}{5}$ and $\sin\theta = \dfrac{4}{5}$ (positive, as $0 < \theta < \pi$).

Double angle: $\cos 2\theta = 2\cos^2\theta - 1 = -\dfrac{7}{25}$, $\sin 2\theta = 2\sin\theta\cos\theta = \dfrac{24}{25}$.

Compound angle: $\cos 3\theta = \cos(2\theta+\theta) = \cos 2\theta\cos\theta - \sin 2\theta\sin\theta = -\dfrac{21}{125} - \dfrac{96}{125} = -\dfrac{117}{125}$ (the same as $4\cos^3\theta - 3\cos\theta$).

Hence the x-coordinate of C is $125 \cdot \left(-\dfrac{117}{125}\right) = -117$.

Final Answer: -117

Question 07

English

Let θ be a real number such that $\tan\theta = \frac{4}{3}$ and $\sin\theta < 0$. Then the value of $\sum_{k=1}^{2026} \cos\left(\theta + \frac{k\pi}{2}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि θ एक वास्तविक संख्या है जिसके लिए $\tan\theta = \frac{4}{3}$ तथा $\sin\theta < 0$ है। तब $\sum_{k=1}^{2026} \cos\left(\theta + \frac{k\pi}{2}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\tan\theta = \frac{4}{3} > 0$ means θ is in the first or third quadrant; $\sin\theta < 0$ selects the third quadrant, where $\sin\theta$ and $\cos\theta$ are both negative.

From $\tan\theta = \frac{4}{3}$: $\sin\theta = -\frac{4}{5}$, $\cos\theta = -\frac{3}{5}$.

Allied angles: $\cos\left(\theta + \frac{\pi}{2}\right) = -\sin\theta$, $\cos(\theta+\pi) = -\cos\theta$, $\cos\left(\theta + \frac{3\pi}{2}\right) = \sin\theta$, $\cos(\theta+2\pi) = \cos\theta$.

The terms repeat with period 4 in k, and each block of four consecutive terms adds to 0.

Since $2026 = 4\cdot506+2$, the first 2024 terms add to 0, and $k = 2025$ ($\equiv 1$), $k = 2026$ ($\equiv 2$) contribute $-\sin\theta$ and $-\cos\theta$.

Sum $= -\sin\theta - \cos\theta = \frac{4}{5} + \frac{3}{5} = \frac{7}{5} = 1.4$.

Final Answer: 1.4

Question 08

English

Let $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$ be such that $\tan\alpha + \tan\beta = 6$ and $\tan(\alpha+\beta) = -\frac{24}{7}$. Then the value of $\cos(\alpha-\beta)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$ इस प्रकार हैं कि $\tan\alpha + \tan\beta = 6$ तथा $\tan(\alpha+\beta) = -\frac{24}{7}$ है। तब $\cos(\alpha-\beta)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\tan(\alpha+\beta) = \dfrac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}$ gives $\dfrac{6}{1-p} = -\dfrac{24}{7}$ with $p = \tan\alpha\tan\beta$, so $1-p = -\dfrac{7}{4}$ and $p = \dfrac{11}{4}$.

(Then $\tan\alpha$, $\tan\beta$ are the roots of $t^2 - 6t + \dfrac{11}{4} = 0$, namely $\dfrac{11}{2}$ and $\dfrac{1}{2}$, both positive, so such α, β exist.)

Compound angles: $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos\alpha\cos\beta(1+p)$ and $\cos(\alpha+\beta) = \cos\alpha\cos\beta(1-p)$.

Dividing: $\dfrac{\cos(\alpha-\beta)}{\cos(\alpha+\beta)} = \dfrac{1+p}{1-p} = \dfrac{15/4}{-7/4} = -\dfrac{15}{7}$.

Since $\alpha+\beta \in (0,\pi)$ and $\tan(\alpha+\beta) < 0$, $\alpha+\beta$ is obtuse: $\cos(\alpha+\beta) = -\dfrac{7}{25}$ (from $\tan = -\dfrac{24}{7}$).

Hence $\cos(\alpha-\beta) = -\dfrac{15}{7}\cdot\left(-\dfrac{7}{25}\right) = \dfrac{3}{5} = 0.6$.

Final Answer: 0.6

Question 09

English

The value of $\sec^3 40^\circ + \sec^3 80^\circ - \sec^3 20^\circ$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\sec^3 40^\circ + \sec^3 80^\circ - \sec^3 20^\circ$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Compound and multiple angles: $\cos 3\theta = \cos(2\theta + \theta) = \cos 2\theta \cos\theta - \sin 2\theta \sin\theta = 4\cos^3\theta - 3\cos\theta$.

For $\theta = 40^\circ, 80^\circ, 160^\circ$: $3\theta = 120^\circ, 240^\circ, 480^\circ$, and each has cosine $-\dfrac{1}{2}$.

So $c = \cos\theta$ satisfies $4c^3 - 3c = -\dfrac{1}{2}$, i.e. $8c^3 - 6c + 1 = 0$. The three values $\cos 40^\circ$, $\cos 80^\circ$, $\cos 160^\circ$ are distinct, so they are exactly the three roots.

Put $u = \dfrac{1}{c} = \sec\theta$: dividing $8c^3 - 6c + 1 = 0$ by c^3 gives $8 - 6u^2 + u^3 = 0$, i.e. $u^3 - 6u^2 + 0 \cdot u + 8 = 0$.

Hence $e_1 = \sum u = 6$, $e_2 = \sum u_i u_j = 0$, $e_3 = u_1 u_2 u_3 = -8$.

Newton's identity: $\sum u^3 = e_1^3 - 3e_1 e_2 + 3e_3 = 216 - 0 - 24 = 192$.

Allied angle: $\cos 160^\circ = -\cos 20^\circ$, so $\sec^3 160^\circ = -\sec^3 20^\circ$. Therefore $\sec^3 40^\circ + \sec^3 80^\circ - \sec^3 20^\circ = 192$.

(Numerically: $2.2245 + 190.9799 - 1.2052 \approx 192.00$.)

Final Answer: 192

Question 10

English

Let $\theta \in \left(0, \dfrac{\pi}{2}\right)$ be such that $\cos 2\theta = \dfrac{1}{3}$. Then the value of $\tan 3\theta \cot\theta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\theta \in \left(0, \dfrac{\pi}{2}\right)$ इस प्रकार है कि $\cos 2\theta = \dfrac{1}{3}$ है। तब $\tan 3\theta \cot\theta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Double angle: $\cos 2\theta = \dfrac{1 - \tan^2\theta}{1 + \tan^2\theta}$. With $\cos 2\theta = \dfrac{1}{3}$: $3(1 - t^2) = 1 + t^2$, so $t^2 = \tan^2\theta = \dfrac{1}{2}$.

Triple angle: $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$, defined because $1 - 3 \cdot \dfrac{1}{2} = -\dfrac{1}{2} \neq 0$.

Dividing by $\tan\theta \neq 0$: $\tan 3\theta \cot\theta = \dfrac{3 - \tan^2\theta}{1 - 3\tan^2\theta} = \dfrac{3 - \frac{1}{2}}{1 - \frac{3}{2}} = \dfrac{5/2}{-1/2} = -5$.

Check by another route: $\dfrac{\sin 3\theta}{\sin\theta} = 1 + 2\cos 2\theta = \dfrac{5}{3}$ and $\dfrac{\cos 3\theta}{\cos\theta} = 2\cos 2\theta - 1 = -\dfrac{1}{3}$, whose ratio is -5.

Final Answer: -5

Question 11

English

Let $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ be such that $\sqrt{2+\sqrt{2+2\cos 4\theta}} = \sqrt{2+\sqrt{3}}$. Then the degree measure of θ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ इस प्रकार है कि $\sqrt{2+\sqrt{2+2\cos 4\theta}} = \sqrt{2+\sqrt{3}}$ है। तब θ का डिग्री माप _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Multiple angle: $1+\cos 4\theta = 2\cos^2 2\theta$, so $2+2\cos 4\theta = 4\cos^2 2\theta$ and $\sqrt{2+2\cos 4\theta} = 2|\cos 2\theta|$.

For $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, $2\theta \in \left(\frac{\pi}{2}, \pi\right)$, so $\cos 2\theta < 0$ and $2|\cos 2\theta| = -2\cos 2\theta$.

Then $2+\sqrt{2+2\cos 4\theta} = 2-2\cos 2\theta = 4\sin^2\theta$, and $\sqrt{4\sin^2\theta} = 2\sin\theta$ because $\sin\theta > 0$.

So the equation is $2\sin\theta = \sqrt{2+\sqrt{3}}$, i.e. $\sin^2\theta = \dfrac{2+\sqrt{3}}{4}$.

Half angle: $\cos^2 15^\circ = \dfrac{1+\cos 30^\circ}{2} = \dfrac{2+\sqrt{3}}{4}$. Hence $\sin\theta = \cos 15^\circ = \sin 75^\circ$.

In $(45^\circ, 90^\circ)$ the sine is one-one, so $\theta = 75^\circ$.

Note: the usual simplification $2\cos\theta$ is valid only when $\cos 2\theta \geq 0$; using it here would give $\theta = 15^\circ$, which is not in the given interval.

Final Answer: 75

Question 12

English

Let $\theta \in \left(\frac{3\pi}{2}, 2\pi\right)$ be such that $7\tan\theta + 24 = 0$. Then the value of $\sin\frac{\theta}{2} + \cos\frac{\theta}{2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\theta \in \left(\frac{3\pi}{2}, 2\pi\right)$ इस प्रकार है कि $7\tan\theta + 24 = 0$ है। तब $\sin\frac{\theta}{2} + \cos\frac{\theta}{2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\tan\theta = -\frac{24}{7}$ with θ in the fourth quadrant, where cosine is positive and sine negative: $\cos\theta = \frac{7}{25}$, $\sin\theta = -\frac{24}{25}$.

$\frac{\theta}{2} \in \left(\frac{3\pi}{4}, \pi\right)$, the second quadrant, so $\sin\frac{\theta}{2} > 0$ and $\cos\frac{\theta}{2} < 0$.

Half-angle formulae: $\sin^2\frac{\theta}{2} = \frac{1-\cos\theta}{2} = \frac{9}{25}$ and $\cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2} = \frac{16}{25}$.

With the signs: $\sin\frac{\theta}{2} = \frac{3}{5}$, $\cos\frac{\theta}{2} = -\frac{4}{5}$.

Hence $\sin\frac{\theta}{2} + \cos\frac{\theta}{2} = -\frac{1}{5} = -0.2$.

Check: $\left(\sin\frac{\theta}{2} + \cos\frac{\theta}{2}\right)^2 = 1 + \sin\theta = \frac{1}{25}$, and the sum is negative because $\left|\cos\frac{\theta}{2}\right| > \sin\frac{\theta}{2}$ in $\left(\frac{3\pi}{4}, \pi\right)$.

Final Answer: -0.2

Question 13

English

Let α and β be real numbers such that $\alpha - \beta = \dfrac{2\pi}{3}$ and $\cos\alpha + \cos\beta = \dfrac{4}{5}$. Then the value of $\cos(\alpha + \beta)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि α तथा β वास्तविक संख्याएँ हैं जिनके लिए $\alpha - \beta = \dfrac{2\pi}{3}$ तथा $\cos\alpha + \cos\beta = \dfrac{4}{5}$ है। तब $\cos(\alpha + \beta)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Sum-to-product formula: $\cos\alpha + \cos\beta = 2\cos\dfrac{\alpha+\beta}{2}\cos\dfrac{\alpha-\beta}{2}$.

Here $\dfrac{\alpha-\beta}{2} = \dfrac{\pi}{3}$, so $\cos\dfrac{\alpha-\beta}{2} = \dfrac{1}{2}$ and $\cos\alpha + \cos\beta = \cos\dfrac{\alpha+\beta}{2}$.

Hence $\cos\dfrac{\alpha+\beta}{2} = \dfrac{4}{5}$.

Then $\cos(\alpha+\beta) = 2\cos^2\dfrac{\alpha+\beta}{2} - 1 = \dfrac{32}{25} - 1 = \dfrac{7}{25} = 0.28$.

The pair (α, β) is not unique (the half angle can be $\pm\cos^{-1}\dfrac{4}{5}$ plus multiples of 2π), but every choice gives the same $\cos(\alpha+\beta)$.

Final Answer: 0.28

Question 14

English

The value of $\operatorname{cosec}^4 18^\circ + \sec^4 36^\circ$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\operatorname{cosec}^4 18^\circ + \sec^4 36^\circ$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Standard values: $\sin 18^\circ = \dfrac{\sqrt{5}-1}{4}$ and $\cos 36^\circ = \dfrac{\sqrt{5}+1}{4}$.

Rationalising: $\operatorname{cosec} 18^\circ = \dfrac{4}{\sqrt{5}-1} = \dfrac{4(\sqrt{5}+1)}{4} = \sqrt{5}+1$ and $\sec 36^\circ = \dfrac{4}{\sqrt{5}+1} = \sqrt{5}-1$.

Let $p = \sqrt{5}+1$ and $q = \sqrt{5}-1$. Then $p^2 = 6+2\sqrt{5}$, $q^2 = 6-2\sqrt{5}$, so $p^2+q^2 = 12$ and $pq = 5-1 = 4$.

$p^4 + q^4 = (p^2+q^2)^2 - 2(pq)^2 = 144 - 32 = 112$.

(Directly: $(6+2\sqrt{5})^2 + (6-2\sqrt{5})^2 = (56+24\sqrt{5}) + (56-24\sqrt{5}) = 112$.)

Final Answer: 112

Question 15

English

The value of $\dfrac{1}{\sin^2 36^\circ \sin^2 72^\circ}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\dfrac{1}{\sin^2 36^\circ \sin^2 72^\circ}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Double angle: $\sin 72^\circ = 2\sin 36^\circ \cos 36^\circ$, so $\sin 36^\circ \sin 72^\circ = 2\sin^2 36^\circ \cos 36^\circ = 2\left(1-\cos^2 36^\circ\right)\cos 36^\circ$.

Non-standard value: $\cos 36^\circ = \dfrac{1+\sqrt{5}}{4}$, hence $\cos^2 36^\circ = \dfrac{6+2\sqrt{5}}{16} = \dfrac{3+\sqrt{5}}{8}$ and $1-\cos^2 36^\circ = \dfrac{5-\sqrt{5}}{8}$.

So $\sin 36^\circ \sin 72^\circ = 2\cdot\dfrac{5-\sqrt{5}}{8}\cdot\dfrac{1+\sqrt{5}}{4} = \dfrac{(5-\sqrt{5})(1+\sqrt{5})}{16} = \dfrac{5+5\sqrt{5}-\sqrt{5}-5}{16} = \dfrac{\sqrt{5}}{4}$.

Therefore $\sin^2 36^\circ \sin^2 72^\circ = \dfrac{5}{16}$ and the required value is $\dfrac{16}{5} = 3.2$.

Final Answer: 3.2

Question 16

English

Let A, B and C be the angles of a triangle such that $\cot A + \cot B + \cot C = \dfrac{5}{2}$. Then the value of $\operatorname{cosec}^2 A + \operatorname{cosec}^2 B + \operatorname{cosec}^2 C$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि A, B तथा C एक त्रिभुज के कोण हैं जिनके लिए $\cot A+\cot B+\cot C = \dfrac{5}{2}$ है। तब $\operatorname{cosec}^2 A+\operatorname{cosec}^2 B+\operatorname{cosec}^2 C$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $A+B+C=\pi$, $\cot(A+B) = \cot(\pi - C) = -\cot C$.

$\cot(A+B) = \dfrac{\cot A\cot B - 1}{\cot A + \cot B}$, so $\cot A\cot B - 1 = -\cot C(\cot A + \cot B)$, i.e. $\cot A\cot B + \cot B\cot C + \cot C\cot A = 1$. (If one angle is a right angle, its cotangent is 0 and the identity still holds.)

Squaring the given sum: $\left(\sum \cot A\right)^2 = \sum \cot^2 A + 2\sum \cot A\cot B$, so $\sum \cot^2 A = \dfrac{25}{4} - 2 = \dfrac{17}{4}$.

Using $\operatorname{cosec}^2 x = 1 + \cot^2 x$: $\sum \operatorname{cosec}^2 A = 3 + \dfrac{17}{4} = \dfrac{29}{4} = 7.25$.

(Such triangles exist: $\cot A + \cot B + \cot C$ takes every value $\geq \sqrt{3}$, and $\dfrac{5}{2} > \sqrt{3}$.)

Final Answer: 7.25

Question 17

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \cos x\,(3\sin x - 4\cos x)$. Then the minimum value of f is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \cos x\,(3\sin x - 4\cos x)$ द्वारा परिभाषित है। तब f का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Expand: $f(x) = 3\sin x\cos x - 4\cos^2 x$.

Double angle: $\sin x\cos x = \dfrac{1}{2}\sin 2x$ and $\cos^2 x = \dfrac{1+\cos 2x}{2}$, so $f(x) = \dfrac{3}{2}\sin 2x - 2\cos 2x - 2$.

For real a, b, $a\sin t + b\cos t$ takes every value in $\left[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}\right]$. Here $\sqrt{\dfrac{9}{4}+4} = \dfrac{5}{2}$.

So f ranges over $\left[-\dfrac{5}{2}-2, \dfrac{5}{2}-2\right] = \left[-\dfrac{9}{2}, \dfrac{1}{2}\right]$, and the minimum $-\dfrac{9}{2} = -4.5$ is attained when $\dfrac{3}{2}\sin 2x - 2\cos 2x = -\dfrac{5}{2}$, i.e. $\sin 2x = -\dfrac{3}{5}$, $\cos 2x = \dfrac{4}{5}$.

Final Answer: -4.5

Question 18

English

Let ABC be a triangle in which $\angle C = \dfrac{\pi}{3}$. Then the maximum value of $7\sin A + 8\sin B$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि ABC एक त्रिभुज है जिसमें $\angle C = \dfrac{\pi}{3}$ है। तब $7\sin A + 8\sin B$ का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The constraint $A + B = \pi - C = \dfrac{2\pi}{3}$ gives $B = \dfrac{2\pi}{3} - A$ with $0 < A < \dfrac{2\pi}{3}$.

$\sin B = \sin\dfrac{2\pi}{3}\cos A - \cos\dfrac{2\pi}{3}\sin A = \dfrac{\sqrt{3}}{2}\cos A + \dfrac{1}{2}\sin A$.

So $7\sin A + 8\sin B = 11\sin A + 4\sqrt{3}\cos A = 13\sin(A+\phi)$, where $\cos\phi = \dfrac{11}{13}$, $\sin\phi = \dfrac{4\sqrt{3}}{13}$.

The value 13 needs $A + \phi = \dfrac{\pi}{2}$, i.e. $\tan A = \cot\phi = \dfrac{11}{4\sqrt{3}} \approx 1.588$, $A \approx 57.8°$, $B \approx 62.2°$. This lies in the allowed range, so the maximum 13 is attained.

Final Answer: 13

Question 19

English

Let x and y be real numbers such that $0 \le y \le x \le \pi$ and $\sin x + \sin y = 1$. Then the greatest possible value of $x - y$, in degrees, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि x तथा y वास्तविक संख्याएँ हैं जिनके लिए $0 \le y \le x \le \pi$ तथा $\sin x + \sin y = 1$ है। तब $x - y$ का अधिकतम सम्भव मान, डिग्री में, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Add the squares: $(\sin x + \sin y)^2 + (\cos x + \cos y)^2 = 2 + 2(\cos x \cos y + \sin x \sin y) = 2 + 2\cos(x - y)$.

With $\sin x + \sin y = 1$: $2 + 2\cos(x - y) = 1 + (\cos x + \cos y)^2 \ge 1$, so $\cos(x - y) \ge -\dfrac{1}{2}$.

Since $0 \le x - y \le \pi$ and cosine decreases on $[0, \pi]$, this means $x - y \le \dfrac{2\pi}{3}$.

Equality needs $\cos x + \cos y = 0$ and $x - y = \dfrac{2\pi}{3}$: $x = \dfrac{5\pi}{6}$, $y = \dfrac{\pi}{6}$ satisfy $0 \le y \le x \le \pi$ and $\sin x + \sin y = \dfrac{1}{2} + \dfrac{1}{2} = 1$.

So the greatest value of $x - y$ is $\dfrac{2\pi}{3}$, that is 120°.

Final Answer: 120

Question 20

English

For $k = 1, 2, \ldots, 11$, let $\theta_k = \dfrac{k\pi}{6}$. Then the value of $\displaystyle\sum_{1 \le j < k \le 11} \cos(\theta_k - \theta_j)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$k = 1, 2, \ldots, 11$ के लिए माना कि $\theta_k = \dfrac{k\pi}{6}$ है। तब $\displaystyle\sum_{1 \le j < k \le 11} \cos(\theta_k - \theta_j)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let S be the required sum, $P = \displaystyle\sum_{k=1}^{11} \cos\theta_k$ and $Q = \displaystyle\sum_{k=1}^{11} \sin\theta_k$.

$P^2 + Q^2 = \sum_k (\cos^2\theta_k + \sin^2\theta_k) + 2\sum_{j<k}(\cos\theta_j \cos\theta_k + \sin\theta_j \sin\theta_k) = 11 + 2\sum_{j<k}\cos(\theta_k - \theta_j) = 11 + 2S$.

Sum of cosines with angles in AP: $\displaystyle\sum_{k=1}^{n} \cos k\alpha = \frac{\sin\frac{n\alpha}{2}}{\sin\frac{\alpha}{2}} \cos\frac{(n+1)\alpha}{2}$, and similarly for sines with $\sin\dfrac{(n+1)\alpha}{2}$.

Here $n = 11$, $\alpha = \dfrac{\pi}{6}$: $\dfrac{\sin\frac{11\pi}{12}}{\sin\frac{\pi}{12}} = 1$ and $\dfrac{(n+1)\alpha}{2} = \pi$, so $P = \cos\pi = -1$ and $Q = \sin\pi = 0$.

Therefore $1 = 11 + 2S$, giving $S = -5$.

(Check by counting: $S = \sum_{d=1}^{10}(11-d)\cos\dfrac{d\pi}{6}$, whose $\sqrt{3}$ terms cancel, also gives -5.)

Final Answer: -5

Question 21

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \cos x \cos\left(x + \dfrac{\pi}{3}\right) + \cos\left(x + \dfrac{2\pi}{3}\right)\cos(x+\pi) + \cos\left(x + \dfrac{4\pi}{3}\right)\cos\left(x + \dfrac{5\pi}{3}\right)$.

Then the value of $f\left(\dfrac{\pi}{7}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \cos x \cos\left(x + \dfrac{\pi}{3}\right) + \cos\left(x + \dfrac{2\pi}{3}\right)\cos(x+\pi) + \cos\left(x + \dfrac{4\pi}{3}\right)\cos\left(x + \dfrac{5\pi}{3}\right)$

द्वारा परिभाषित है। तब $f\left(\dfrac{\pi}{7}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write the three products as $\cos\phi_m \cos\left(\phi_m + \dfrac{\pi}{3}\right)$ with $\phi_m = x + \dfrac{2m\pi}{3}$, $m = 0, 1, 2$ (indeed $\phi_1 + \dfrac{\pi}{3} = x + \pi$ and $\phi_2 + \dfrac{\pi}{3} = x + \dfrac{5\pi}{3}$).

Product to sum: $2\cos P \cos Q = \cos(P+Q) + \cos(P-Q)$ gives $\cos\phi_m \cos\left(\phi_m + \dfrac{\pi}{3}\right) = \dfrac{1}{2}\cos\left(2\phi_m + \dfrac{\pi}{3}\right) + \dfrac{1}{4}$.

So $f(x) = \dfrac{3}{4} + \dfrac{1}{2}\displaystyle\sum_{m=0}^{2}\cos\left(2x + \frac{\pi}{3} + \frac{4m\pi}{3}\right)$.

Angles in AP with first term $a = 2x + \dfrac{\pi}{3}$, difference $d = \dfrac{4\pi}{3}$, $n = 3$ terms: the sum is $\dfrac{\sin\frac{nd}{2}}{\sin\frac{d}{2}}\cos\left(a + \dfrac{(n-1)d}{2}\right)$ with $\sin\dfrac{nd}{2} = \sin 2\pi = 0$ and $\sin\dfrac{d}{2} = \sin\dfrac{2\pi}{3} \neq 0$, so it is 0.

Hence $f(x) = \dfrac{3}{4}$ for every real x, and $f\left(\dfrac{\pi}{7}\right) = 0.75$.

(At $x = \dfrac{\pi}{7}$ no factor has a standard value, so direct evaluation is not practical.)

Final Answer: 0.75

Question 22

English

The value of $\sum_{k=1}^{5} \tan\frac{k\pi}{7}\,\tan\frac{(k+1)\pi}{7}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\sum_{k=1}^{5} \tan\frac{k\pi}{7}\,\tan\frac{(k+1)\pi}{7}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $A = \frac{(k+1)\pi}{7}$, $B = \frac{k\pi}{7}$: $\tan(A-B) = \frac{\tan A - \tan B}{1+\tan A\tan B}$ with $A - B = \frac{\pi}{7}$.

So $1+\tan A\tan B = \cot\frac{\pi}{7}(\tan A - \tan B)$, i.e. $\tan\frac{k\pi}{7}\tan\frac{(k+1)\pi}{7} = \cot\frac{\pi}{7}\left(\tan\frac{(k+1)\pi}{7} - \tan\frac{k\pi}{7}\right) - 1$.

None of $\frac{k\pi}{7}$, $k = 1,\ldots,6$, is an odd multiple of $\frac{\pi}{2}$, so every tangent is defined.

Adding for $k = 1$ to 5, the bracket telescopes: $\sum_{k=1}^{5} \tan\frac{k\pi}{7}\tan\frac{(k+1)\pi}{7} = \cot\frac{\pi}{7}\left(\tan\frac{6\pi}{7} - \tan\frac{\pi}{7}\right) - 5$.

Allied angle: $\tan\frac{6\pi}{7} = \tan\left(\pi - \frac{\pi}{7}\right) = -\tan\frac{\pi}{7}$.

Hence the sum is $\cot\frac{\pi}{7}\cdot\left(-2\tan\frac{\pi}{7}\right) - 5 = -2 - 5 = -7$.

Final Answer: -7

Question 23

English

Let θ be a real number with $\sin\theta \neq -1$ and $\cos\theta \neq -1$, and let $x = \dfrac{\cos\theta}{1+\sin\theta}$ and $y = \dfrac{\sin\theta}{1+\cos\theta}$. If $x = 4$, then the value of y is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि θ एक वास्तविक संख्या है जिसके लिए $\sin\theta \neq -1$ तथा $\cos\theta \neq -1$ है, तथा $x = \dfrac{\cos\theta}{1+\sin\theta}$ और $y = \dfrac{\sin\theta}{1+\cos\theta}$ है। यदि $x = 4$ है, तो y का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $c = \cos\theta$, $s = \sin\theta$; the denominators $1+s$ and $1+c$ are non-zero.

$$x + y + xy = \frac{c}{1+s} + \frac{s}{1+c} + \frac{cs}{(1+s)(1+c)} = \frac{c(1+c)+s(1+s)+cs}{(1+s)(1+c)}.$$

Numerator $= c + s + (c^2+s^2) + cs = 1 + c + s + cs = (1+c)(1+s)$, using $c^2+s^2 = 1$.

So θ is eliminated: $x + y + xy = 1$ for every admissible θ.

(Equivalently $y = \tan\dfrac{\theta}{2}$ and $x = \tan\left(\dfrac{\pi}{4} - \dfrac{\theta}{2}\right) = \dfrac{1-y}{1+y}$.)

With $x = 4$: $4 + y + 4y = 1$, so $y = -\dfrac{3}{5} = -0.6$.

Existence check: $\cos\theta = 4(1+\sin\theta)$ with $\sin\theta = -\dfrac{15}{17}$, $\cos\theta = \dfrac{8}{17}$ gives $x = 4$ and $y = \dfrac{-15/17}{25/17} = -0.6$.

Final Answer: -0.6

Question 24

English

Let $\alpha \in \left(0, \frac{\pi}{2}\right)$ with $\tan\alpha = \frac{3}{4}$. For $\theta \in \mathbb{R}$, let $x = 5\cos(\theta+\alpha)$ and $y = 5\cos(\theta-\alpha)$. Eliminating θ, the relation between x and y can be written as $px^2 + qxy + py^2 = r$, where p, q, r are integers having no common factor other than 1 and $p > 0$. Then the value of r is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\alpha \in \left(0, \frac{\pi}{2}\right)$ तथा $\tan\alpha = \frac{3}{4}$ है। $\theta \in \mathbb{R}$ के लिए माना कि $x = 5\cos(\theta+\alpha)$ तथा $y = 5\cos(\theta-\alpha)$ है। θ का विलोपन करने पर x तथा y के बीच का सम्बन्ध $px^2 + qxy + py^2 = r$ के रूप में लिखा जा सकता है, जहाँ p, q, r ऐसे पूर्णांक हैं जिनका 1 के अतिरिक्त कोई उभयनिष्ठ गुणनखण्ड नहीं है तथा $p > 0$ है। तब r का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

From $\tan\alpha = \frac{3}{4}$ with α acute: $\cos\alpha = \frac{4}{5}$, $\sin\alpha = \frac{3}{5}$.

Compound angles: $x = 5(\cos\theta\cos\alpha - \sin\theta\sin\alpha) = 4\cos\theta - 3\sin\theta$ and $y = 5(\cos\theta\cos\alpha + \sin\theta\sin\alpha) = 4\cos\theta + 3\sin\theta$.

Adding and subtracting: $x + y = 8\cos\theta$ and $y - x = 6\sin\theta$.

Eliminate θ by $\cos^2\theta + \sin^2\theta = 1$: $\frac{(x+y)^2}{64} + \frac{(x-y)^2}{36} = 1$.

Multiply by 576 (the LCM of 64 and 36): $9(x+y)^2 + 16(x-y)^2 = 576$, i.e. $25x^2 + 25y^2 + (18-32)xy = 576$, so $25x^2 - 14xy + 25y^2 = 576$.

The coefficients 25, -14, 576 have no common factor ($25 = 5^2$, $14 = 2 \cdot 7$), and $p = 25 > 0$; the form is unique under these conditions. Hence $r = 576$.

(Check: $\theta = 0$ gives $x = y = 4$ and $25 \cdot 16 - 14 \cdot 16 + 25 \cdot 16 = 576$.)

Final Answer: 576